Exploratory Data Analysis (EDA)

Rx Drug Marketing Mix Modeling (MMIX)

Business Context

Pharmaceutical firms compete within therapeutic classes, where multiple brands seek to influence prescriptions through two primary channels:

- **Detailing**: physician-targeted promotion
- **DTCA**: direct-to-consumer advertising

The commercial mechanism is:

> Marketing investment → influences physicians and patients → drives prescriptions → generates sales

Business Objective

The goal of this analysis is to understand how marketing investments translate into sales outcomes across brands and therapeutic classes, and to identify patterns that should guide model design and budget decisions.

Core Business Questions

1. Which brands dominate portfolio revenue, and how concentrated is the portfolio?
2. How has portfolio performance evolved over time?
3. How are detailing and DTCA budgets allocated across brands and classes?
4. Which channel appears more efficient before formal modeling?
5. Is there evidence of diminishing returns to marketing spend?
6. Do classes and brands differ in baseline sales and marketing sensitivity?
7. Are there structurally different marketing regimes, especially around zero DTCA?
8. What does the panel structure imply for modeling and validation?

Scope of Analysis

This review encompasses an 11-year historical dataset, capturing the following parameters:

- Observation Period: 2013–2024
- Therapeutic Classes: PPI (Proton Pump Inhibitors), SSRI (Antidepressants), Statin (Cholesterol)
- Portfolio Size: 22 distinct brands
- Key Metrics Evaluated:
  - Top-line Sales in Billions of Dollars
  - Physician Detailing Spend in Millions of Dollars
  - Direct-to-Consumer Advertising (DTCA) Spend ($ Millions)

In [43]:
```
print("\n[EDA] Importing libraries...")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")
pd.set_option("display.max_columns", 100)
pd.set_option("display.float_format", lambda x: f"{x:,.4f}")

print("[EDA] Libraries loaded successfully.")
```

```
[EDA] Importing libraries...
[EDA] Libraries loaded successfully.
```

In [44]:
```
print("\n[EDA] Loading dataset and standardizing column names...")

df = pd.read_excel("data/MMM_Drug_Data.xlsx")

df = df.rename(columns={
    "Class": "class",
    "Agent": "agent",
    "Year": "year",
    "Detailing": "detailing",
    "Direct-to-Consumer Advertising (DTCA)": "direct_to_consumer_ad",
    "Actual Sales": "actual_sales"
})

print("[EDA] Dataset loaded successfully.")
display(df.head())
```

```
[EDA] Loading dataset and standardizing column names...
[EDA] Dataset loaded successfully.
```

	class	agent	year	detailing	direct_to_consumer_ad	actual_sales
0	PPI	Prevacid	2013	43.8805	0.0000	2.9083
1	PPI	Prilosec	2013	41.6480	0.0000	14.7235
2	SSRI	Luvox	2013	22.3667	0.0000	0.9120
3	SSRI	Paxil	2013	57.4608	0.0000	11.2677
4	SSRI	Prozac	2013	84.5333	0.0000	20.2600

In [45]:
```
print("\n[EDA] Running initial data checks...")

print("Shape:", df.shape)

print("\nColumn types:")
display(df.dtypes.to_frame("dtype"))

print("\nMissing values:")
display(df.isna().sum().to_frame("missing_count"))

print("\nYears covered:", df["year"].min(), "to", df["year"].max())
print("Number of classes:", df["class"].nunique())
print("Number of agents:", df["agent"].nunique())

print("\n[Business Insight]")
print("The dataset covers 12 years, 22 brands, and 3 therapeutic classes with no missing valu
es.")
print("This supports both cross-sectional and panel-style analysis.")
```

```
[EDA] Running initial data checks...
Shape: (207, 6)

Column types:
```

	dtype
class	str
agent	str
year	int64
detailing	float64
direct_to_consumer_ad	float64
actual_sales	float64

```
Missing values:
```

	missing_count
class	0
agent	0
year	0
detailing	0
direct_to_consumer_ad	0
actual_sales	0

```
Years covered: 2013 to 2024
Number of classes: 3
Number of agents: 22

[Business Insight]
The dataset covers 12 years, 22 brands, and 3 therapeutic classes with no missing values.
This supports both cross-sectional and panel-style analysis.
```

In [46]:
```
print("\n[EDA] Checking panel structure (agent × year)...")

panel = pd.crosstab(df["agent"], df["year"])

n_agents = df["agent"].nunique()
n_years = df["year"].nunique()
expected_rows = n_agents * n_years
actual_rows = len(df)

print(f"Expected rows in a balanced panel: {expected_rows}")
print(f"Actual rows observed: {actual_rows}")
print(f"Missing agent-year combinations: {expected_rows - actual_rows}")

display(panel)

test_agents = set(df[df["year"] >= 2023]["agent"].unique())
train_agents = set(df[df["year"] <= 2020]["agent"].unique())

print("\n[EDA] Train/test overlap check for future time-based modeling...")
print(f"Agents in test (2023-2024) but not in train (<=2020): {test_agents - train_agents or
'None'}")
print(f"Agents in train (<=2020) but not in test (2023-2024): {train_agents - test_agents or
'None'}")

print("\n[Business Insight]")
print("The portfolio is an unbalanced panel, which is realistic for markets where brands laun
ch, mature, or exit over time.")
print("This matters for model validation because time-based splits must avoid cold-start bran
ds in the test period.")
```

```
[EDA] Checking panel structure (agent × year)...
Expected rows in a balanced panel: 264
Actual rows observed: 207
Missing agent-year combinations: 57
```

year	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
agent												
Aciphex	0	0	0	1	1	1	1	1	1	1	1	1
Altoprev	0	0	0	0	0	0	1	1	1	1	1	1
Celexa	0	0	1	1	1	1	1	1	1	1	1	1
Crestor	0	0	0	0	0	0	0	1	1	1	1	1
Lescol	1	1	1	1	1	1	1	1	1	1	1	1
Lescol XL	0	0	0	0	1	1	1	1	1	1	1	1
Lexapro	0	0	0	0	0	0	1	1	1	1	1	1
Lipitor	0	1	1	1	1	1	1	1	1	1	1	1
Luvox	1	1	1	1	1	1	1	1	1	1	1	1
Mevacor	1	1	1	1	1	1	1	1	1	1	1	1
Nexium	0	0	0	0	0	1	1	1	1	1	1	1
Paxil	1	1	1	1	1	1	1	1	1	1	1	1
Paxil CR	0	0	0	0	0	0	1	1	1	1	1	1
Pexeva	0	0	0	0	0	0	0	0	1	1	1	1
Pravachol	1	1	1	1	1	1	1	1	1	1	1	1
Prevacid	1	1	1	1	1	1	1	1	1	1	1	1
Prilosec	1	1	1	1	1	1	1	1	1	1	1	1
Protonix	0	0	0	0	1	1	1	1	1	1	1	1
Prozac	1	1	1	1	1	1	1	1	1	1	1	1
Prozac Weekly	0	0	0	0	0	1	1	1	1	1	1	1
Zocor	1	1	1	1	1	1	1	1	1	1	1	1
Zoloft	1	1	1	1	1	1	1	1	1	1	1	1

```
[EDA] Train/test overlap check for future time-based modeling...
Agents in test (2023-2024) but not in train (<=2020): {'  Pexeva'}
Agents in train (<=2020) but not in test (2023-2024): None

[Business Insight]
The portfolio is an unbalanced panel, which is realistic for markets where brands launch, mat
ure, or exit over time.
This matters for model validation because time-based splits must avoid cold-start brands in t
he test period.
```

In [47]:
```
print("\n[EDA] Generating summary statistics for key commercial variables...")

summary = df[["actual_sales", "detailing", "direct_to_consumer_ad"]].describe()

summary_readable = summary.copy()
summary_readable["actual_sales"] = summary["actual_sales"].apply(lambda x: f"${x:,.2f}B")
summary_readable["detailing"] = summary["detailing"].apply(lambda x: f"${x:,.2f}M")
summary_readable["direct_to_consumer_ad"] = summary["direct_to_consumer_ad"].apply(lambda x:
f"${x:,.2f}M")

summary_readable = summary_readable.reset_index().rename(columns={"index": "Metric"})
display(summary_readable)

print("\n[Business Insight]")
print("Sales and marketing spend exhibit high variance and dispersion, suggesting uneven scal
e across brands. This motivates deeper analysis of concentration and distribution.")
print("Both detailing and DTCA vary widely across the portfolio, indicating materially differ
ent channel strategies by brand and class.")
```

```
[EDA] Generating summary statistics for key commercial variables...
```

	Metric	actual_sales	detailing	direct_to_consumer_ad
0	count	$207.00B	$207.00M	$207.00M
1	mean	$11.67B	$56.03M	$26.32M
2	std	$12.97B	$54.92M	$49.30M
3	min	$0.00B	$0.00M	$0.00M
4	25%	$0.99B	$3.86M	$0.00M
5	50%	$7.53B	$44.97M	$0.00M
6	75%	$19.61B	$85.12M	$39.36M
7	max	$63.22B	$243.33M	$240.35M

```
[Business Insight]
Sales and marketing spend exhibit high variance and dispersion, suggesting uneven scale acros
s brands. This motivates deeper analysis of concentration and distribution.
Both detailing and DTCA vary widely across the portfolio, indicating materially different cha
nnel strategies by brand and class.
```

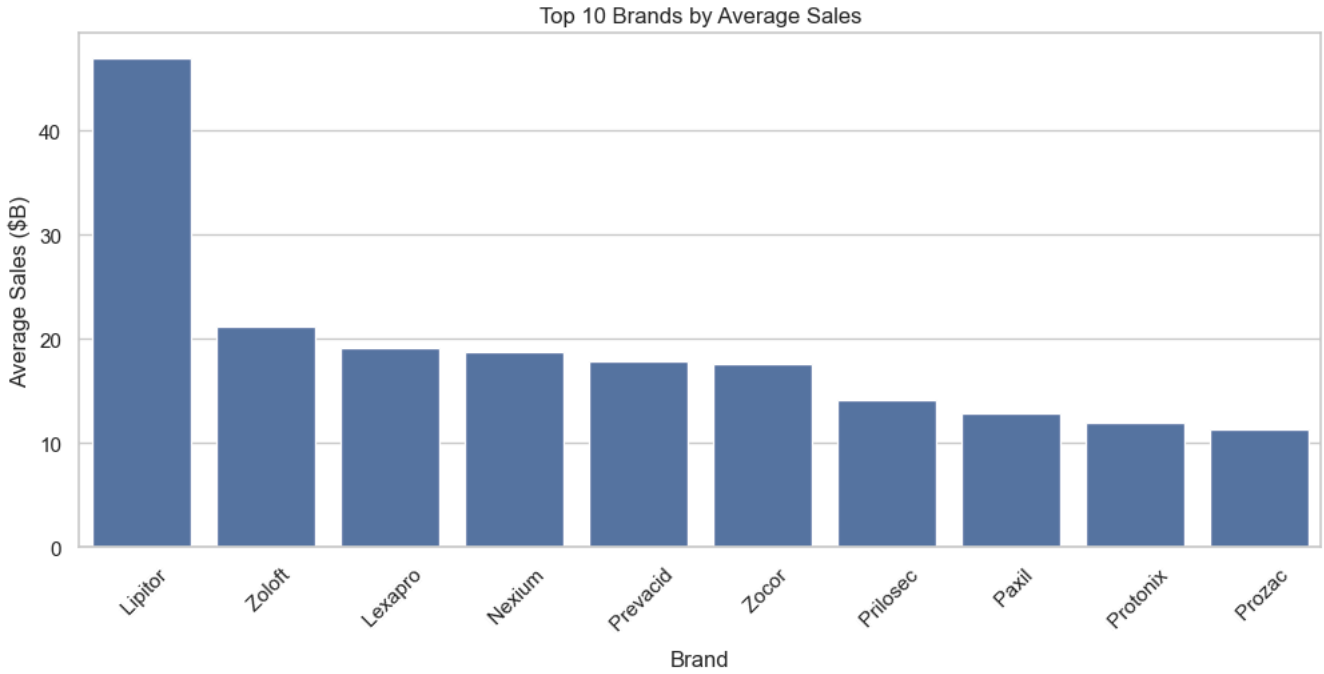
1. Revenue Concentration

A first strategic question is whether sales are diversified or concentrated in a few blockbuster brands.

In [48]:
```
print("\n[EDA] Ranking brands by average sales contribution...")

brand_sales = (
    df.groupby("agent", as_index=False)["actual_sales"]
    .mean()
    .sort_values("actual_sales", ascending=False)
)

brand_sales["sales_share_pct"] = 100 * brand_sales["actual_sales"] / brand_sales["actual_sale
s"].sum()
brand_sales["cumulative_share_pct"] = brand_sales["sales_share_pct"].cumsum()

display(brand_sales.head(10))

plt.figure(figsize=(12, 5))
sns.barplot(data=brand_sales.head(10), x="agent", y="actual_sales")
plt.title("Top 10 Brands by Average Sales")
plt.xlabel("Brand")
plt.ylabel("Average Sales ($B)")
plt.xticks(rotation=45)
plt.show()

top_5_share = brand_sales.head(5)["sales_share_pct"].sum()

print("\n[Business Insight]")
print(f"The top 5 brands account for approximately {top_5_share:.1f}% of portfolio average sa
les.")
print("This indicates meaningful concentration risk and implies that a few brands can disprop
ortionately influence pooled model estimates.")
```

```
[EDA] Ranking brands by average sales contribution...
```

	agent	actual_sales	sales_share_pct	cumulative_share_pct
7	Lipitor	47.0275	19.7929	19.7929
21	Zoloft	21.1441	8.8991	28.6920
6	Lexapro	19.1402	8.0557	36.7477
10	Nexium	18.7587	7.8951	44.6428
15	Prevacid	17.8747	7.5231	52.1659
20	Zocor	17.6050	7.4096	59.5754
16	Prilosec	14.1557	5.9578	65.5333
11	Paxil	12.8184	5.3950	70.9283
17	Protonix	12.0249	5.0610	75.9893
18	Prozac	11.2791	4.7471	80.7364




```
[Business Insight]
The top 5 brands account for approximately 52.2% of portfolio average sales.
This indicates meaningful concentration risk and implies that a few brands can disproportiona
tely influence pooled model estimates.
```

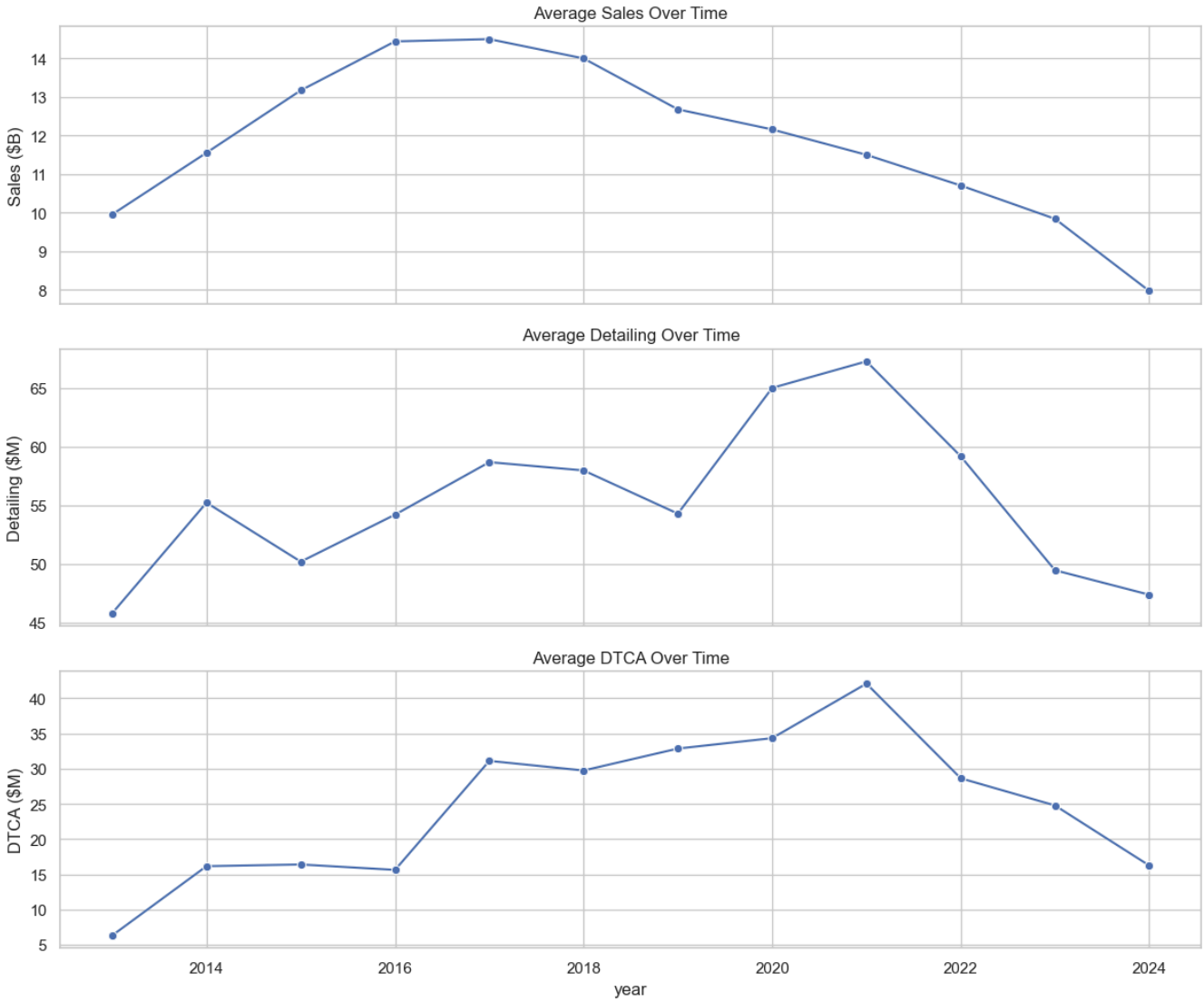
2. Growth Trajectory

Before asking which marketing works, we need to understand whether the portfolio is expanding, maturing, or declining.

In [49]:
```
print("\n[EDA] Analyzing portfolio growth trajectory over time...")

yearly = (
    df.groupby("year", as_index=False)[["actual_sales", "detailing", "direct_to_consumer_a
d"]]
    .mean()
)

yearly["sales_yoy_pct"] = yearly["actual_sales"].pct_change() * 100
yearly["detailing_yoy_pct"] = yearly["detailing"].pct_change() * 100
yearly["dtca_yoy_pct"] = yearly["direct_to_consumer_ad"].pct_change() * 100

display(yearly)

fig, axes = plt.subplots(3, 1, figsize=(12, 10), sharex=True)

sns.lineplot(data=yearly, x="year", y="actual_sales", marker="o", ax=axes[0])
axes[0].set_title("Average Sales Over Time")
axes[0].set_ylabel("Sales ($B)")

sns.lineplot(data=yearly, x="year", y="detailing", marker="o", ax=axes[1])
axes[1].set_title("Average Detailing Over Time")
axes[1].set_ylabel("Detailing ($M)")

sns.lineplot(data=yearly, x="year", y="direct_to_consumer_ad", marker="o", ax=axes[2])
axes[2].set_title("Average DTCA Over Time")
axes[2].set_ylabel("DTCA ($M)")

plt.tight_layout()
plt.show()

print("\n[Business Insight]")
print("The portfolio shows a non-linear lifecycle, with sales rising through the middle years
and declining toward 2024.")
print("This implies that a simple linear time trend is unlikely to capture the commercial dyn
amics adequately.")
```

```
[EDA] Analyzing portfolio growth trajectory over time...
```

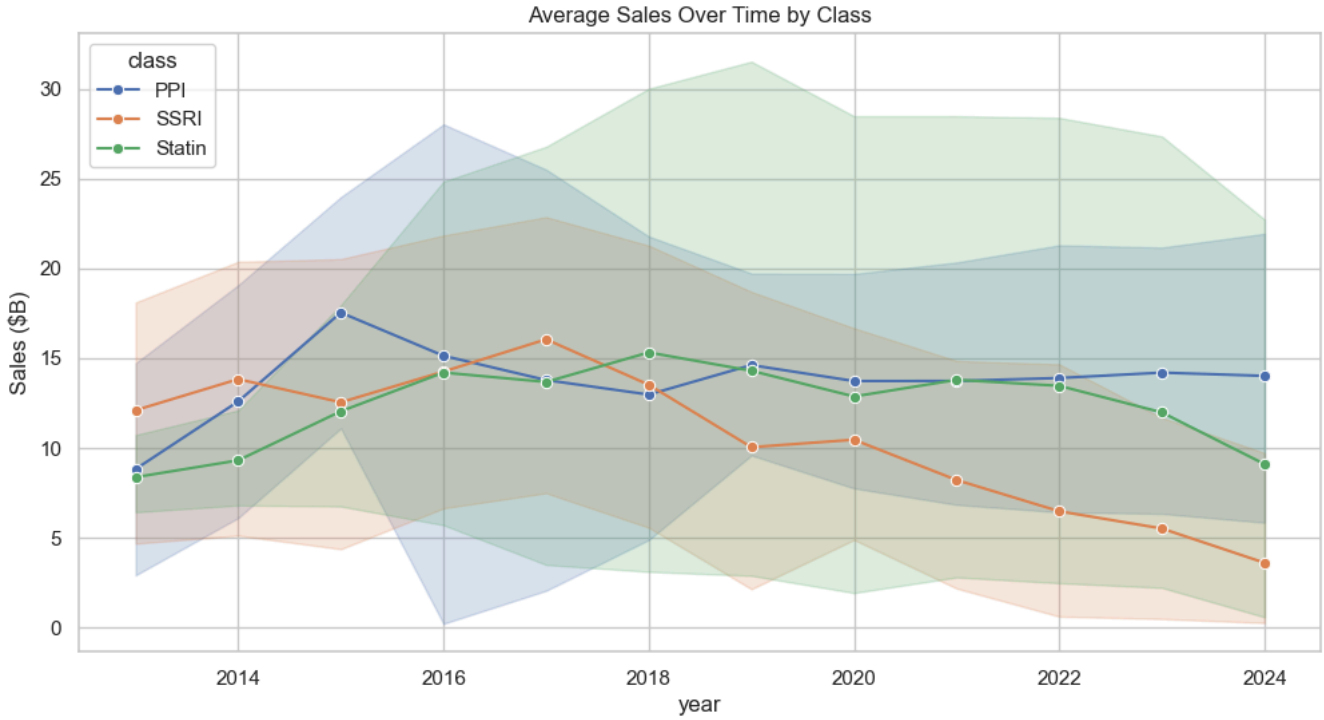
	year	actual_sales	detailing	direct_to_consumer_ad	sales_yoy_pct	detailing_yoy_pct	dtca_yoy_pct
0	2013	9.9486	45.7758	6.3109	NaN	NaN	NaN
1	2014	11.5475	55.2130	16.1484	16.0714	20.6161	155.8812
2	2015	13.1646	50.1752	16.3910	14.0041	-9.1242	1.5025
3	2016	14.4348	54.1879	15.6212	9.6486	7.9974	-4.6969
4	2017	14.4936	58.6714	31.1021	0.4072	8.2740	99.1020
5	2018	13.9928	57.9632	29.7359	-3.4555	-1.2071	-4.3926
6	2019	12.6725	54.2737	32.8522	-9.4353	-6.3652	10.4801
7	2020	12.1514	65.0097	34.3327	-4.1124	19.7812	4.5066
8	2021	11.4936	67.2739	42.0747	-5.4132	3.4829	22.5497
9	2022	10.6978	59.1707	28.6292	-6.9237	-12.0451	-31.9561
10	2023	9.8339	49.4446	24.7726	-8.0753	-16.4374	-13.4710
11	2024	7.9631	47.3541	16.2398	-19.0238	-4.2278	-34.4443




```
[Business Insight]
The portfolio shows a non-linear lifecycle, with sales rising through the middle years and de
clining toward 2024.
This implies that a simple linear time trend is unlikely to capture the commercial dynamics a
dequately.
```

In [50]:
```
print("\n[EDA] Comparing class-level sales trends over time...")

plt.figure(figsize=(12, 6))
sns.lineplot(data=df, x="year", y="actual_sales", hue="class", marker="o")
plt.title("Average Sales Over Time by Class")
plt.ylabel("Sales ($B)")
plt.show()

print("\n[Business Insight]")
print("Classes do not follow identical trajectories, suggesting that both market maturity and
class-specific dynamics shape demand.")
print("This supports the later use of class controls and year effects in the parametric mode
l.")
```

```
[EDA] Comparing class-level sales trends over time...
```




```
[Business Insight]
Classes do not follow identical trajectories, suggesting that both market maturity and class-
specific dynamics shape demand.
This supports the later use of class controls and year effects in the parametric model.
```

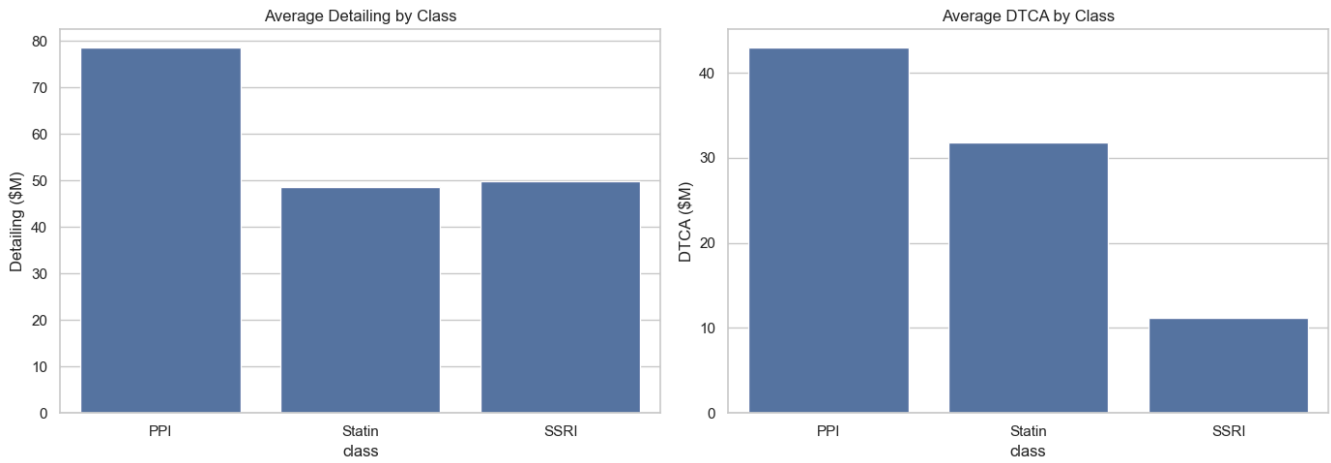
3. Budget Allocation

The next question is how the firm allocates detailing and DTCA budgets across classes and brands.

In [51]:
```
print("\n[EDA] Summarizing average budget allocation by class...")

class_spend = (
    df.groupby("class", as_index=False)[["actual_sales", "detailing", "direct_to_consumer_a
d"]]
    .mean()
    .sort_values("actual_sales", ascending=False)
)

display(class_spend)

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sns.barplot(data=class_spend, x="class", y="detailing", ax=axes[0])
axes[0].set_title("Average Detailing by Class")
axes[0].set_ylabel("Detailing ($M)")

sns.barplot(data=class_spend, x="class", y="direct_to_consumer_ad", ax=axes[1])
axes[1].set_title("Average DTCA by Class")
axes[1].set_ylabel("DTCA ($M)")

plt.tight_layout()
plt.show()

print("\n[Business Insight]")
print("The company does not deploy the same marketing mix across classes.")
print("This suggests that channel strategy is likely class-contingent, not one-size-fits-al
l.")
```

```
[EDA] Summarizing average budget allocation by class...
```

	class	actual_sales	detailing	direct_to_consumer_ad
0	PPI	13.8355	78.5221	42.9419
2	Statin	12.4964	48.5592	31.7618
1	SSRI	9.5923	49.9067	11.2310




```
[Business Insight]
The company does not deploy the same marketing mix across classes.
This suggests that channel strategy is likely class-contingent, not one-size-fits-all.
```

In [52]:
```
print("\n[EDA] Identifying brands with the highest marketing investment...")

brand_spend = (
    df.groupby("agent", as_index=False)[["actual_sales", "detailing", "direct_to_consumer_a
d"]]
    .mean()
    .sort_values("detailing", ascending=False)
)

display(brand_spend.head(10))

print("\n[Business Insight]")
print("Some brands are detailing-heavy, some are DTCA-heavy, and some invest materially in bo
th channels.")
print("This variation provides the commercial heterogeneity that the later model must explai
n.")
```

```
[EDA] Identifying brands with the highest marketing investment...
```

	agent	actual_sales	detailing	direct_to_consumer_ad
6	Lexapro	19.1402	192.7008	0.7499
10	Nexium	18.7587	153.0828	182.8566
3	Crestor	7.9550	142.5837	119.2919
7	Lipitor	47.0275	124.0093	80.7884
15	Prevacid	17.8747	92.0651	38.7137
17	Protonix	12.0249	83.2735	3.3244
21	Zoloft	21.1441	80.4162	30.3664
20	Zocor	17.6050	70.3491	54.1879
0	Aciphex	5.8032	66.7499	1.1366
12	Paxil CR	4.8374	53.8639	24.0865

```
[Business Insight]
Some brands are detailing-heavy, some are DTCA-heavy, and some invest materially in both chan
nels.
This variation provides the commercial heterogeneity that the later model must explain.
```

4. Baseline Efficiency

Before formal modeling, it is useful to compute simple descriptive efficiency ratios: how much revenue is associated with each unit of spend?

In [53]:
```
print("\n[EDA] Computing descriptive baseline efficiency by channel...")

efficiency = df.copy()
efficiency["sales_per_detailing"] = np.where(
    efficiency["detailing"] > 0,
    efficiency["actual_sales"] / efficiency["detailing"],
    np.nan
)
efficiency["sales_per_dtca"] = np.where(
    efficiency["direct_to_consumer_ad"] > 0,
    efficiency["actual_sales"] / efficiency["direct_to_consumer_ad"],
    np.nan
)

efficiency_summary = (
    efficiency.groupby("class")[["sales_per_detailing", "sales_per_dtca"]]
    .mean()
    .sort_values("sales_per_detailing", ascending=False)
)

display(efficiency_summary)

avg_eff = efficiency[["sales_per_detailing", "sales_per_dtca"]].mean()
display(avg_eff.to_frame("portfolio_avg_efficiency"))

print("\n[Business Insight]")
print("These ratios are highly sensitive to low spend levels and structural zeros, particular
ly in DTCA. As a result, they should be interpreted cautiously and should not be used as a ba
sis for strategic decisions without controlled modeling.")
```

```
[EDA] Computing descriptive baseline efficiency by channel...
```

class	sales_per_detailing	sales_per_dtca
Statin	0.5474	59.6006
PPI	0.3913	335.6168
SSRI	0.3591	1,979.3762

	portfolio_avg_efficiency
sales_per_detailing	0.4387
sales_per_dtca	739.4620

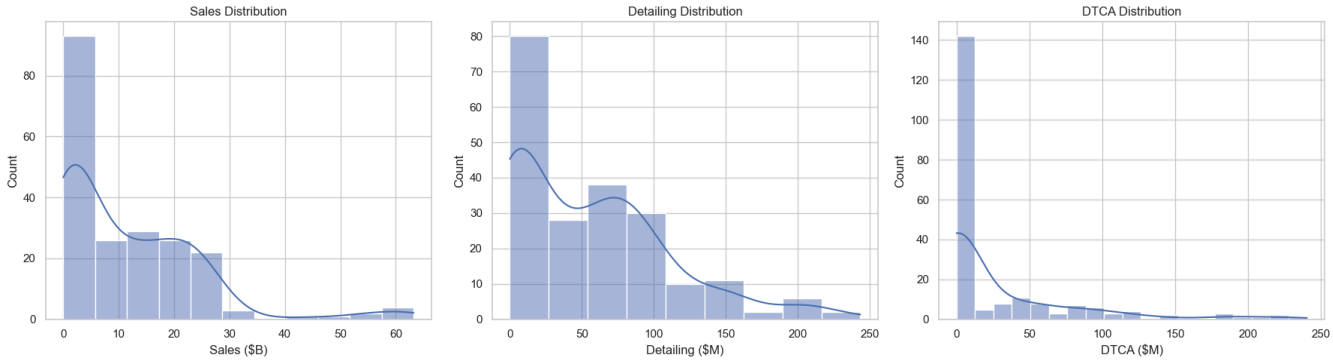
```
[Business Insight]
These ratios are highly sensitive to low spend levels and structural zeros, particularly in D
TCA. As a result, they should be interpreted cautiously and should not be used as a basis for
strategic decisions without controlled modeling.
```

5. Distribution Shape and Market Structure

We now examine whether sales and spend are broadly distributed or concentrated in a small number of observations.

In [54]:
```
print("\n[EDA] Visualizing raw distributions of sales and marketing spend...")

fig, axes = plt.subplots(1, 3, figsize=(18, 5))

sns.histplot(df["actual_sales"], kde=True, ax=axes[0])
axes[0].set_title("Sales Distribution")
axes[0].set_xlabel("Sales ($B)")

sns.histplot(df["detailing"], kde=True, ax=axes[1])
axes[1].set_title("Detailing Distribution")
axes[1].set_xlabel("Detailing ($M)")

sns.histplot(df["direct_to_consumer_ad"], kde=True, ax=axes[2])
axes[2].set_title("DTCA Distribution")
axes[2].set_xlabel("DTCA ($M)")

plt.tight_layout()
plt.show()

print("\n[Business Insight]")
print("All three variables are strongly right-skewed, which is typical in blockbuster-driven
markets.")
print("A small number of brands or campaigns account for a disproportionate share of scale.")
```

```
[EDA] Visualizing raw distributions of sales and marketing spend...
```




```
[Business Insight]
All three variables are strongly right-skewed, which is typical in blockbuster-driven market
s.
A small number of brands or campaigns account for a disproportionate share of scale.
```

In [55]:
```
print("\n[EDA] Quantifying skewness in raw variables...")

skewness_raw = df[["actual_sales", "detailing", "direct_to_consumer_ad"]].skew()
display(skewness_raw.to_frame("skewness"))

print("\n[Business Insight]")
print("Positive skew confirms that raw-scale relationships are likely to be influenced by ext
reme observations.")
print("This motivates transformed specifications before parametric modeling.")
```

```
[EDA] Quantifying skewness in raw variables...
```

	skewness
actual_sales	1.7127
detailing	1.0358
direct_to_consumer_ad	2.2988

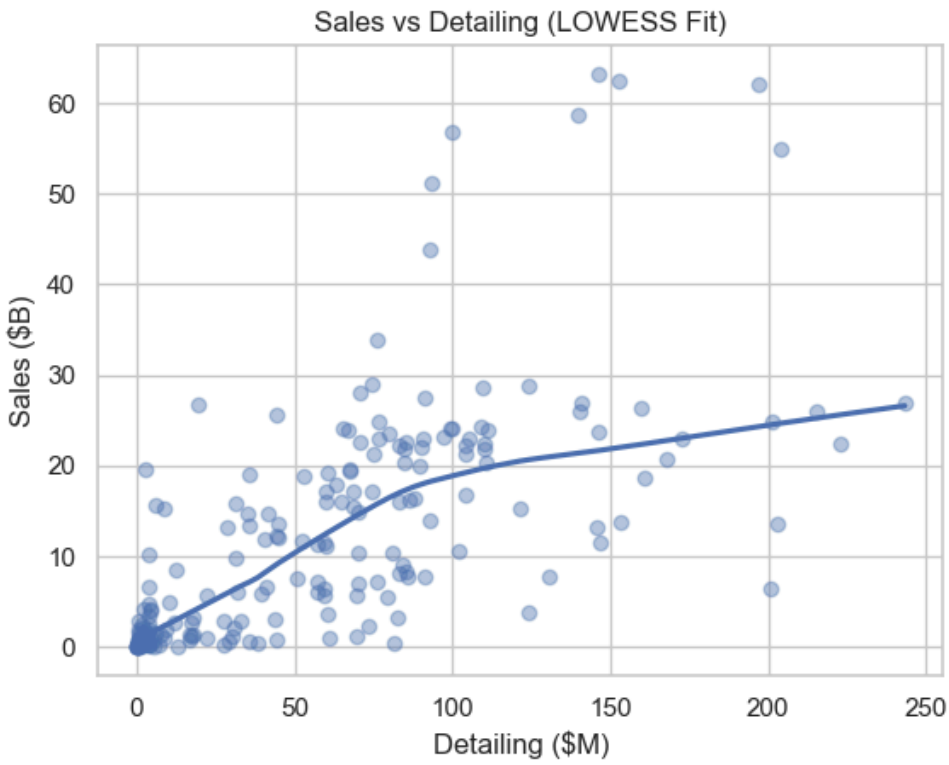
```
[Business Insight]
Positive skew confirms that raw-scale relationships are likely to be influenced by extreme ob
servations.
This motivates transformed specifications before parametric modeling.
```

6. Spend Saturation and Diminishing Returns

The next business question is whether more spend always produces proportionally more sales.

In [56]:
```
print("\n[EDA] Testing for diminishing returns in detailing using LOWESS...")

sns.regplot(
    data=df,
    x="detailing",
    y="actual_sales",
    lowess=True,
    scatter_kws={"alpha": 0.4}
)

plt.title("Sales vs Detailing (LOWESS Fit)")
plt.xlabel("Detailing ($M)")
plt.ylabel("Sales ($B)")
plt.show()

print("\n[Business Insight]")
print("The curve flattens at higher spend levels, which is consistent with diminishing margin
al returns to detailing.")
```

```
[EDA] Testing for diminishing returns in detailing using LOWESS...
```




```
[Business Insight]
The curve flattens at higher spend levels, which is consistent with diminishing marginal retu
rns to detailing.
```

In [57]:
```
print("\n[EDA] Approximating where detailing saturation begins...")

df["detailing_bucket"] = pd.qcut(df["detailing"], q=10, duplicates="drop")
bucket_sales = df.groupby("detailing_bucket", observed=False)["actual_sales"].mean().to_frame
("avg_sales")

display(bucket_sales)

print("\n[Business Insight]")
print("Sales increase with higher spend, but at a decreasing rate, reinforcing the presence o
f diminishing returns..")
print("That pattern suggests spend saturation rather than constant incremental returns.")
```

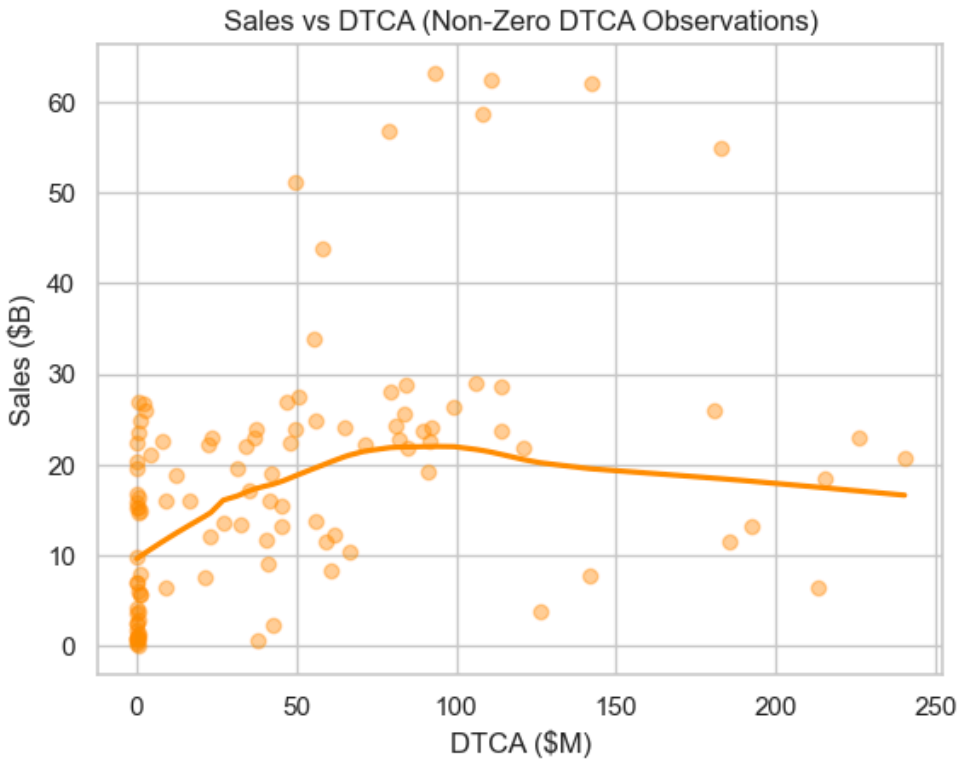
```
[EDA] Approximating where detailing saturation begins...
```

detailing_bucket	avg_sales
(-0.001, 0.639]	0.3719
(0.639, 2.96]	2.0414
(2.96, 6.907]	2.9686
(6.907, 28.631]	4.5371
(28.631, 44.969]	8.5018
(44.969, 67.258]	12.4589
(67.258, 80.046]	16.5847
(80.046, 92.864]	16.3139
(92.864, 134.211]	23.0129
(134.211, 243.33]	29.7583

```
[Business Insight]
Sales increase with higher spend, but at a decreasing rate, reinforcing the presence of dimin
ishing returns..
That pattern suggests spend saturation rather than constant incremental returns.
```

In [58]:
```
print("\n[EDA] Testing for diminishing returns in DTCA among active advertisers...")

non_zero_dtca = df[df["direct_to_consumer_ad"] > 0]

sns.regplot(
    data=non_zero_dtca,
    x="direct_to_consumer_ad",
    y="actual_sales",
    lowess=True,
    scatter_kws={"alpha": 0.4},
    color="darkorange"
)

plt.title("Sales vs DTCA (Non-Zero DTCA Observations)")
plt.xlabel("DTCA ($M)")
plt.ylabel("Sales ($B)")
plt.show()

print("\n[Business Insight]")
print("Among brands that actually advertise to consumers, the relationship appears positive b
ut not strictly linear.")
print("This supports testing transformed or nonlinear response functions later.")
```

```
[EDA] Testing for diminishing returns in DTCA among active advertisers...
```




```
[Business Insight]
Among brands that actually advertise to consumers, the relationship appears positive but not
strictly linear.
This supports testing transformed or nonlinear response functions later.
```

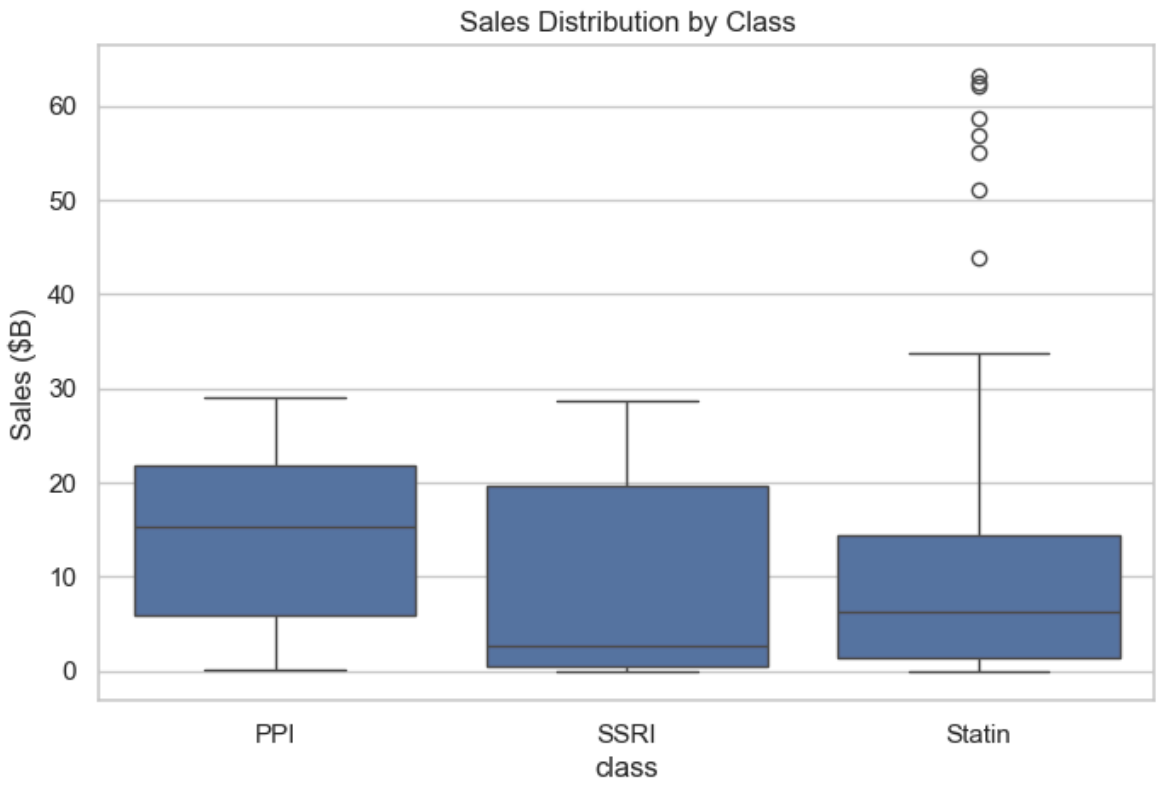
7. Class and Brand Differences in Baseline Sales and Response

A key strategic issue is whether classes and brands differ structurally in both scale and likely marketing response.

In [59]:
```
print("\n[EDA] Comparing baseline sales by class...")

avg_sales_by_class = (
    df.groupby("class", as_index=False)["actual_sales"]
    .mean()
    .sort_values("actual_sales", ascending=False)
)

display(avg_sales_by_class)

plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x="class", y="actual_sales")
plt.title("Sales Distribution by Class")
plt.ylabel("Sales ($B)")
plt.show()

print("\n[Business Insight]")
print("Classes differ in both average sales and dispersion.")
print("This indicates that class membership captures meaningful structural differences in dem
and and competitive conditions.")
```

```
[EDA] Comparing baseline sales by class...
```

	class	actual_sales
0	PPI	13.8355
2	Statin	12.4964
1	SSRI	9.5923

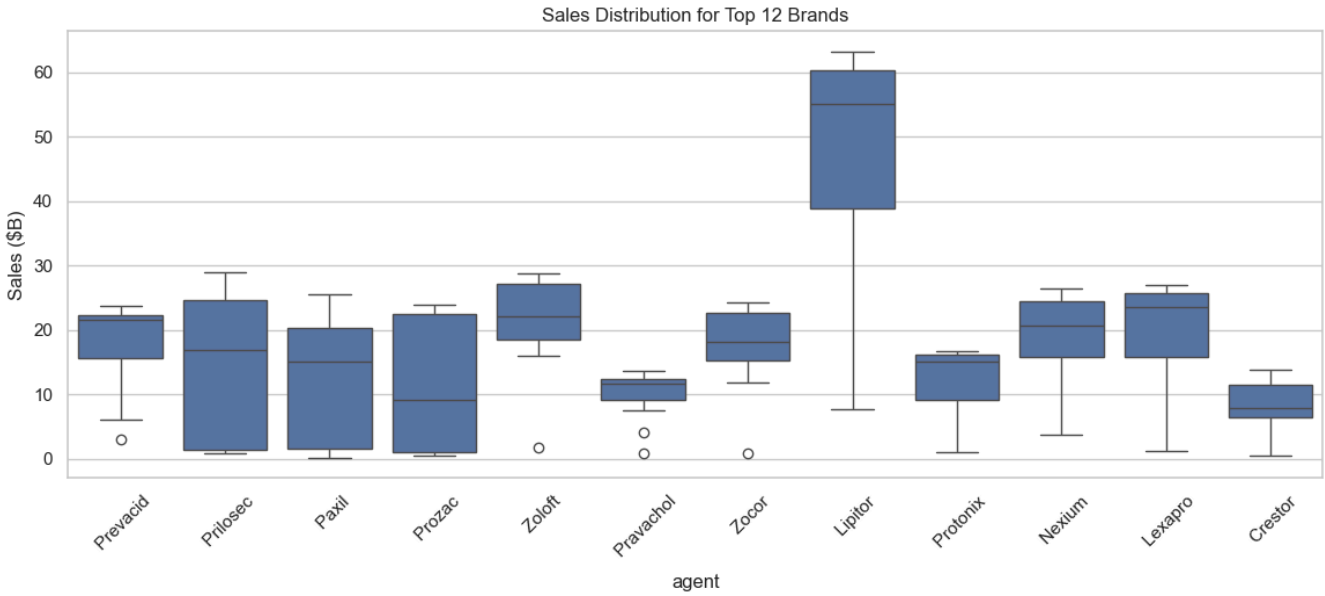



```
[Business Insight]
Classes differ in both average sales and dispersion.
This indicates that class membership captures meaningful structural differences in demand and
competitive conditions.
```

In [60]:
```
print("\n[EDA] Comparing baseline sales by brand...")

avg_sales_by_agent = (
    df.groupby("agent", as_index=False)["actual_sales"]
    .mean()
    .sort_values("actual_sales", ascending=False)
)

display(avg_sales_by_agent.head(15))

top_agents = avg_sales_by_agent.head(12)["agent"].tolist()

plt.figure(figsize=(14, 5))
sns.boxplot(data=df[df["agent"].isin(top_agents)], x="agent", y="actual_sales")
plt.title("Sales Distribution for Top 12 Brands")
plt.ylabel("Sales ($B)")
plt.xticks(rotation=45)
plt.show()

print("\n[Business Insight]")
print("Brand-level baseline differences are substantial and extend well beyond class-level ef
fects.")
print("That means agent effects should be treated as a core modeling feature, not a minor con
trol.")
```

```
[EDA] Comparing baseline sales by brand...
```

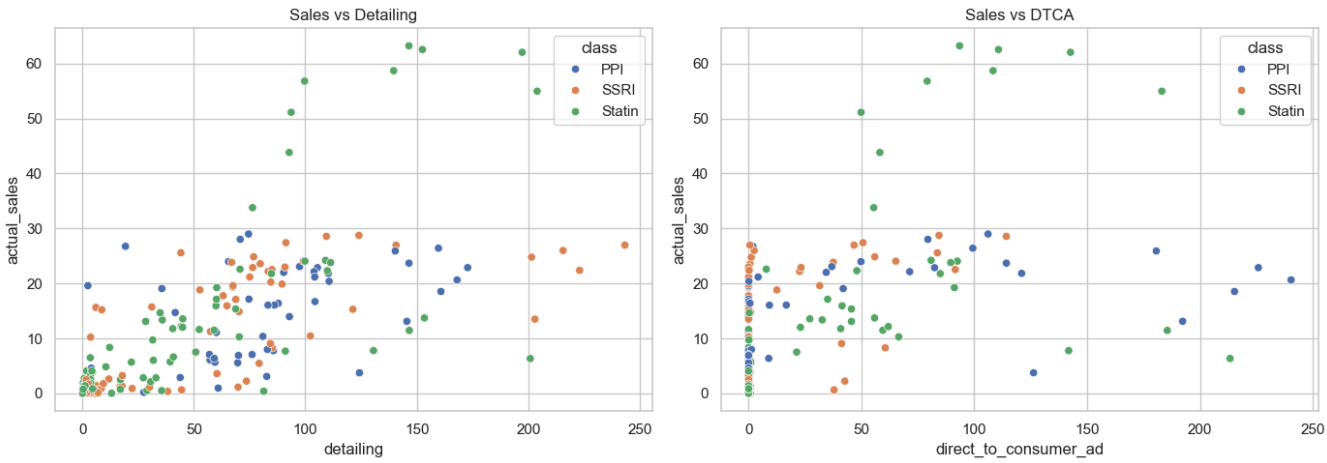
	agent	actual_sales
7	Lipitor	47.0275
21	Zoloft	21.1441
6	Lexapro	19.1402
10	Nexium	18.7587
15	Prevacid	17.8747
20	Zocor	17.6050
16	Prilosec	14.1557
11	Paxil	12.8184
17	Protonix	12.0249
18	Prozac	11.2791
14	Pravachol	9.9861
3	Crestor	7.9550
2	Celexa	7.8978
0	Aciphex	5.8032
12	Paxil CR	4.8374




```
[Business Insight]
Brand-level baseline differences are substantial and extend well beyond class-level effects.
That means agent effects should be treated as a core modeling feature, not a minor control.
```

In [61]:
```
print("\n[EDA] Examining raw response relationships by class...")

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sns.scatterplot(data=df, x="detailing", y="actual_sales", hue="class", ax=axes[0])
axes[0].set_title("Sales vs Detailing")

sns.scatterplot(data=df, x="direct_to_consumer_ad", y="actual_sales", hue="class", ax=axes
[1])
axes[1].set_title("Sales vs DTCA")

plt.tight_layout()
plt.show()

print("\n[Business Insight]")
print("The raw relationship between marketing spend and sales is noisy, reinforcing the need
for transformed specifications..")
```

```
[EDA] Examining raw response relationships by class...
```




```
[Business Insight]
The raw relationship between marketing spend and sales is noisy, reinforcing the need for tra
nsformed specifications..
```

8. Structural Zero-DTCA Regime

A large share of observations show exactly zero DTCA, which may reflect a distinct strategic regime rather than just low spend.

In [62]:
```
print("\n[EDA] Measuring the prevalence and structure of zero DTCA...")

df["dtca_zero"] = (df["direct_to_consumer_ad"] == 0).astype(int)

zero_dtca = df["dtca_zero"].sum()
total = len(df)

print(f"Zero DTCA observations: {zero_dtca}")
print(f"Total observations: {total}")
print(f"Share of zero-DTCA observations: {zero_dtca / total:.2%}")

print("\nZero DTCA by class:")
display(pd.crosstab(df["class"], df["dtca_zero"]))

print("\nZero DTCA by year:")
display(pd.crosstab(df["year"], df["dtca_zero"]))

print("\n[Business Insight]")
print("A meaningful share of brands operate without DTCA, implying a dual-regime strategy: ph
ysician-led brands versus consumer-advertised brands.")
print("That distinction may matter enough to model explicitly later.")
```

```
[EDA] Measuring the prevalence and structure of zero DTCA...
Zero DTCA observations: 98
Total observations: 207
Share of zero-DTCA observations: 47.34%

Zero DTCA by class:
```

dtca_zero	0	1
class		
PPI	32	16
SSRI	34	47
Statin	43	35

```
Zero DTCA by year:
```

dtca_zero	0	1
year		
2013	3	7
2014	5	6
2015	6	6
2016	6	7
2017	7	8
2018	12	5
2019	12	8
2020	10	11
2021	13	9
2022	13	9
2023	10	12
2024	12	10

```
[Business Insight]
A meaningful share of brands operate without DTCA, implying a dual-regime strategy: physician
-led brands versus consumer-advertised brands.
That distinction may matter enough to model explicitly later.
```

9. Log Transformations for Parametric Modeling

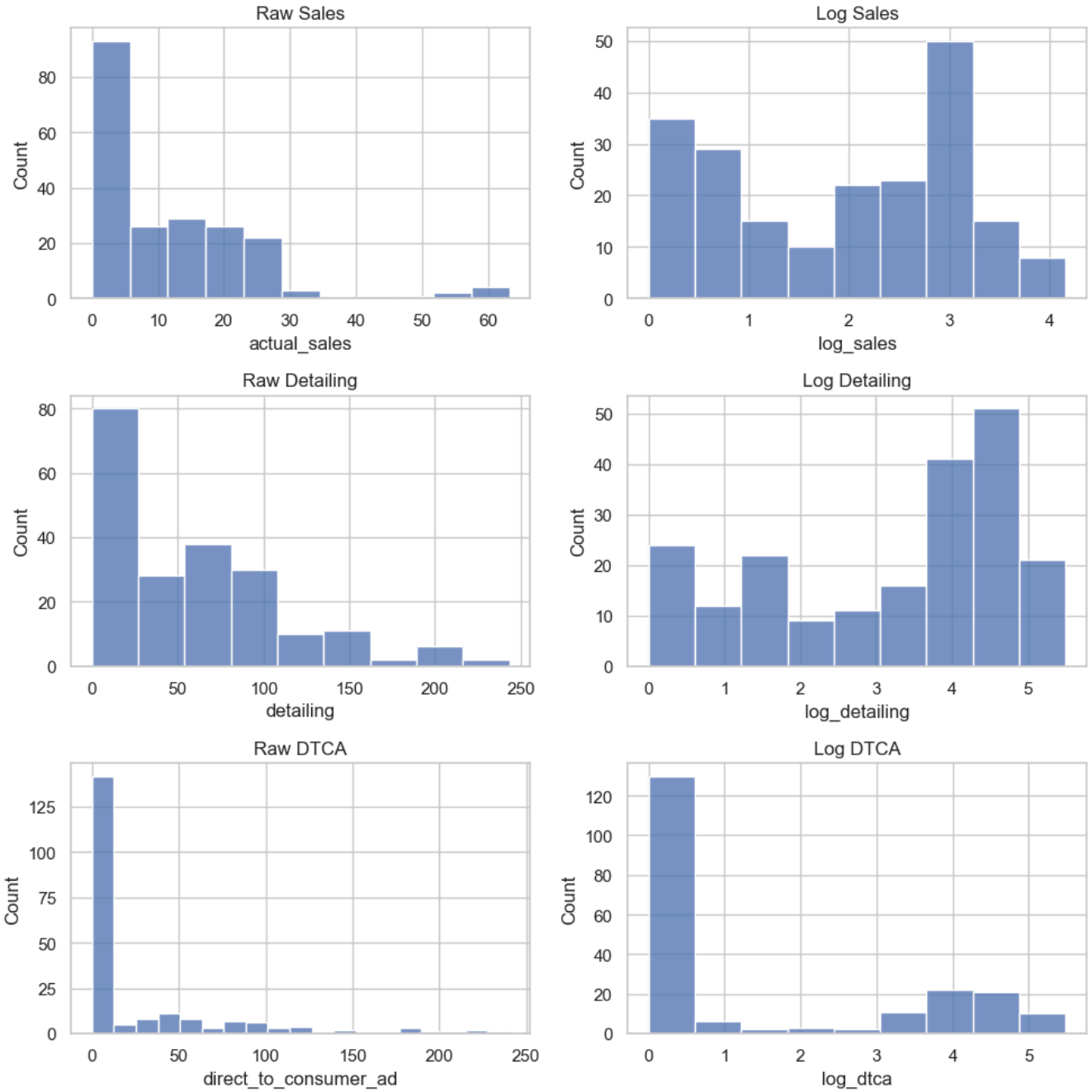
Because the raw variables are highly skewed and the commercial response appears multiplicative, we prepare log variables for the next stage.

In [63]:
```
print("\n[EDA] Creating log-transformed variables for next-stage modeling...")

df["log_sales"] = np.log1p(df["actual_sales"])
df["log_detailing"] = np.log1p(df["detailing"])
df["log_dtca"] = np.log1p(df["direct_to_consumer_ad"])

print("[EDA] Log variables created: log_sales, log_detailing, log_dtca")
```

```
[EDA] Creating log-transformed variables for next-stage modeling...
[EDA] Log variables created: log_sales, log_detailing, log_dtca
```

In [64]:
```
print("\n[EDA] Comparing raw and log-transformed distributions...")

fig, axes = plt.subplots(3, 2, figsize=(10, 10))

sns.histplot(df["actual_sales"], ax=axes[0, 0])
axes[0, 0].set_title("Raw Sales")

sns.histplot(df["log_sales"], ax=axes[0, 1])
axes[0, 1].set_title("Log Sales")

sns.histplot(df["detailing"], ax=axes[1, 0])
axes[1, 0].set_title("Raw Detailing")

sns.histplot(df["log_detailing"], ax=axes[1, 1])
axes[1, 1].set_title("Log Detailing")

sns.histplot(df["direct_to_consumer_ad"], ax=axes[2, 0])
axes[2, 0].set_title("Raw DTCA")

sns.histplot(df["log_dtca"], ax=axes[2, 1])
axes[2, 1].set_title("Log DTCA")

plt.tight_layout()
plt.show()

print("\n[Business Insight]")
print("The log transform compresses extreme values and produces more stable distributions, es
pecially for sales and detailing.")
print("DTCA improves as well, though the mass at zero remains a structural feature.")
```

```
[EDA] Comparing raw and log-transformed distributions...
```




```
[Business Insight]
The log transform compresses extreme values and produces more stable distributions, especiall
y for sales and detailing.
DTCA improves as well, though the mass at zero remains a structural feature.
```

In [65]:
```
print("\n[EDA] Comparing skewness before and after log transformation...")

skew_before = df[["actual_sales", "detailing", "direct_to_consumer_ad"]].skew()
skew_after = df[["log_sales", "log_detailing", "log_dtca"]].skew()

skew_compare = pd.DataFrame({
    "Before": skew_before,
    "After": skew_after
})

display(skew_compare)

print("\n[Business Insight]")
print("Skewness falls materially after log transformation, which strengthens the case for log
-based parametric models.")
```

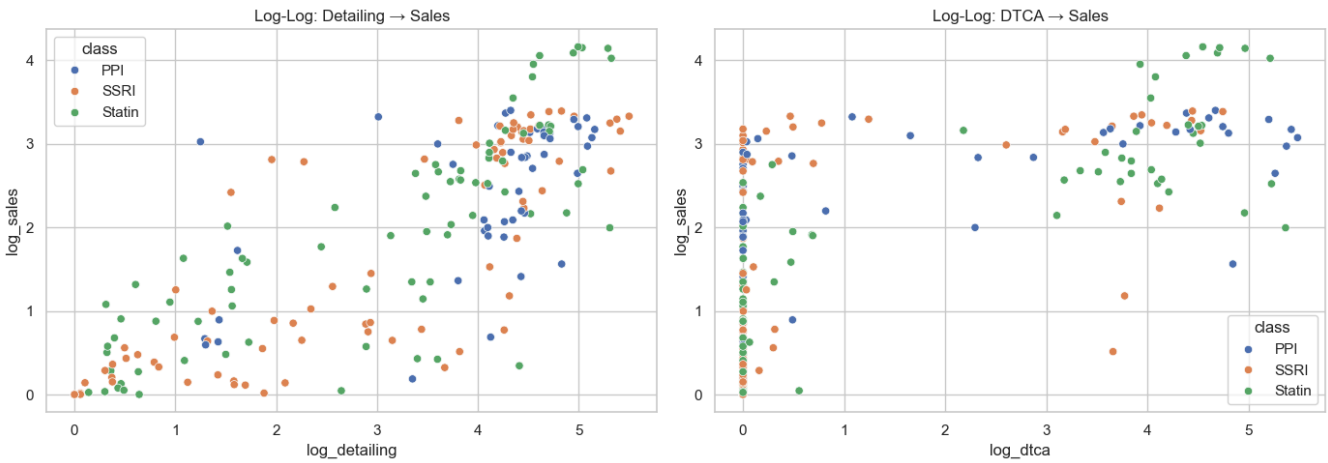
```
[EDA] Comparing skewness before and after log transformation...
```

	Before	After
actual_sales	1.7127	NaN
detailing	1.0358	NaN
direct_to_consumer_ad	2.2988	NaN
log_detailing	NaN	-0.6077
log_dtca	NaN	0.8257
log_sales	NaN	-0.1557

```
[Business Insight]
Skewness falls materially after log transformation, which strengthens the case for log-based
parametric models.
```

In [66]:
```
print("\n[EDA] Evaluating log-log relationships...")

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sns.scatterplot(data=df, x="log_detailing", y="log_sales", hue="class", ax=axes[0])
axes[0].set_title("Log-Log: Detailing → Sales")

sns.scatterplot(data=df, x="log_dtca", y="log_sales", hue="class", ax=axes[1])
axes[1].set_title("Log-Log: DTCA → Sales")

plt.tight_layout()
plt.show()

print("\n[Business Insight]")
print("Relationships become more linear in log space, supporting elasticity-style interpretat
ion in the next stage.")
```

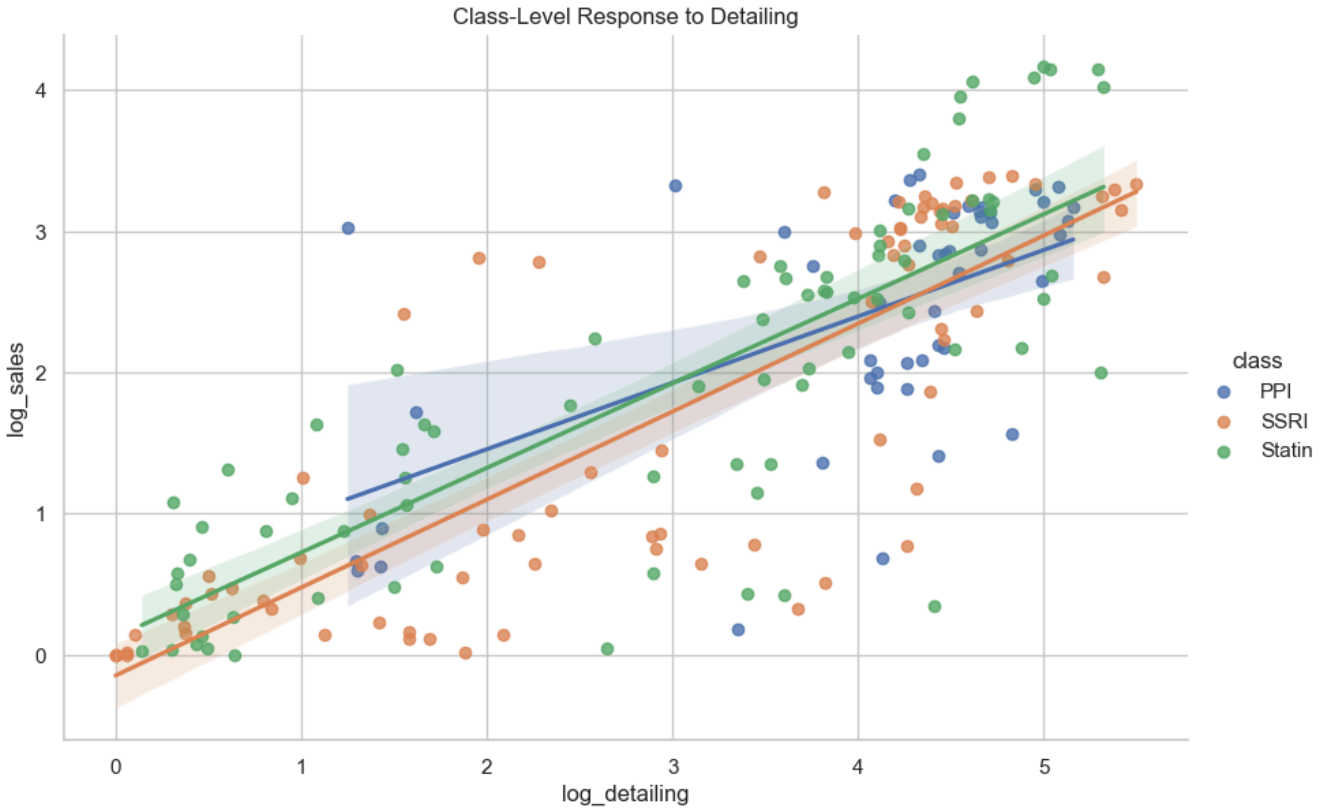
```
[EDA] Evaluating log-log relationships...
```




```
[Business Insight]
Relationships become more linear in log space, supporting elasticity-style interpretation in
the next stage.
```

In [67]:
```
print("\n[EDA] Comparing class-level marketing sensitivity in log space...")

sns.lmplot(
    data=df,
    x="log_detailing",
    y="log_sales",
    hue="class",
    height=6,
    aspect=1.5
)

plt.title("Class-Level Response to Detailing")
plt.show()

print("\n[Business Insight]")
print("Different slopes by class suggest different marketing sensitivities across therapeutic categories.")
print("This supports testing class interaction terms in the parametric model.")
```

```
[EDA] Comparing class-level marketing sensitivity in log space...
```



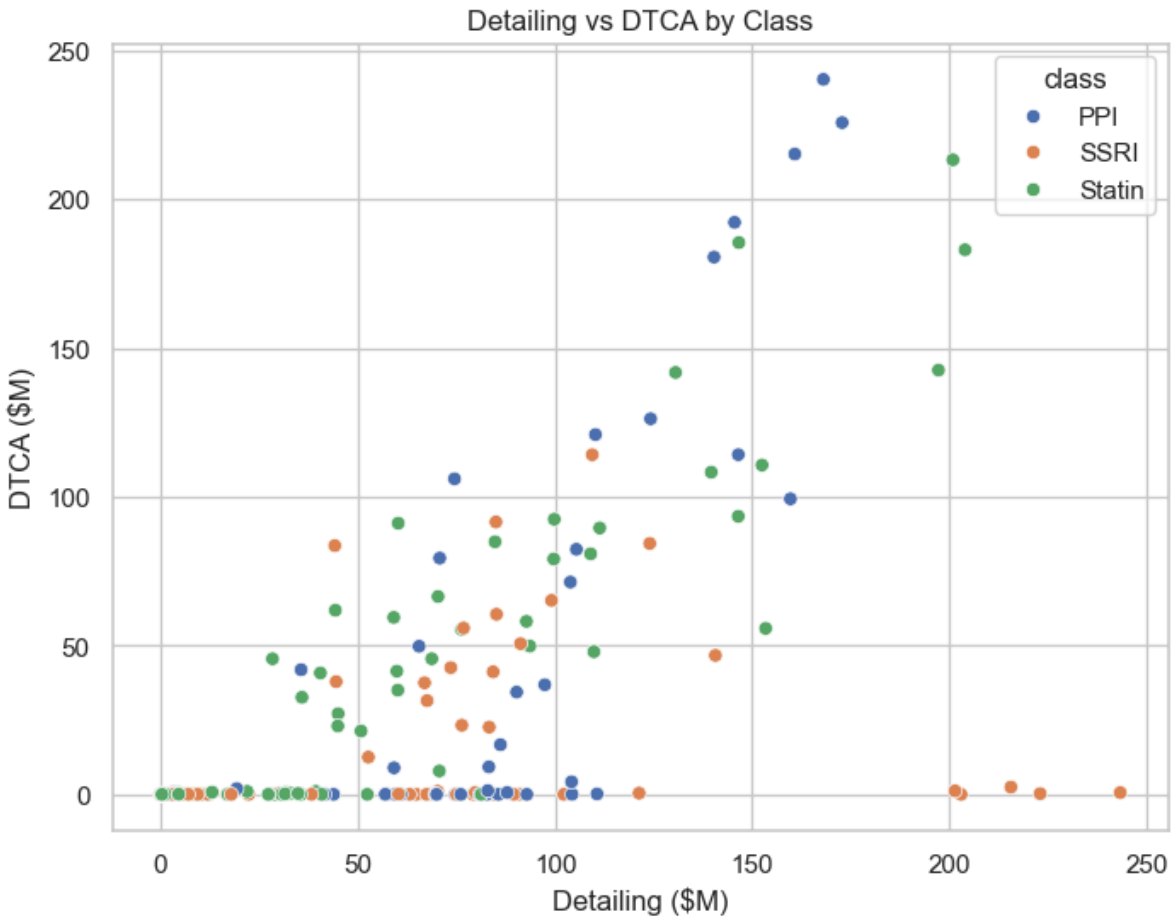

```
[Business Insight]
Different slopes by class suggest different marketing sensitivities across therapeutic catego
ries.
This supports testing class interaction terms in the parametric model.
```

10. Cross-Channel Strategy and Collinearity

Do brands tend to invest in both channels together, or do they specialize?

In [68]:
```
print("\n[EDA] Examining the relationship between detailing and DTCA...")

plt.figure(figsize=(8, 6))
sns.scatterplot(
    data=df,
    x="detailing",
    y="direct_to_consumer_ad",
    hue="class"
)
plt.title("Detailing vs DTCA by Class")
plt.xlabel("Detailing ($M)")
plt.ylabel("DTCA ($M)")
plt.show()

print("\n[Business Insight]")
print("Some classes appear to invest jointly across both channels, while others skew more hea
vily toward one.")
print("This informs both channel strategy interpretation and later multicollinearity check
s.")
```

```
[EDA] Examining the relationship between detailing and DTCA...
```

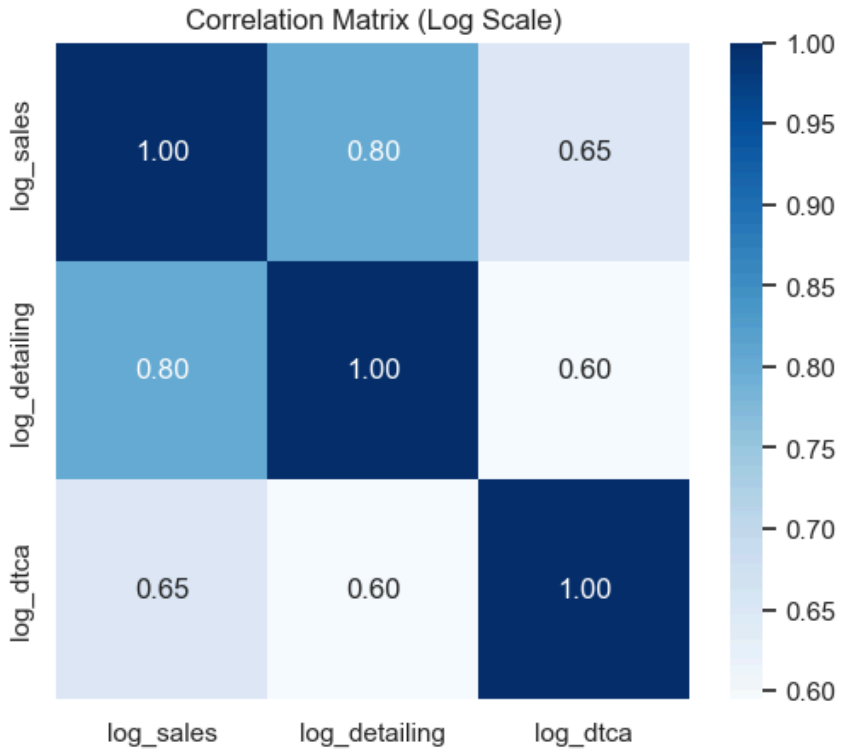



```
[Business Insight]
Some classes appear to invest jointly across both channels, while others skew more heavily to
ward one.
This informs both channel strategy interpretation and later multicollinearity checks.
```

In [69]:
```
print("\n[EDA] Calculating correlation matrix for log-transformed variables...")

corr = df[["log_sales", "log_detailing", "log_dtca"]].corr()

display(corr)

plt.figure(figsize=(6, 5))
sns.heatmap(corr, annot=True, cmap="Blues", fmt=".2f")
plt.title("Correlation Matrix (Log Scale)")
plt.show()

print("\n[Business Insight]")
print("Detailing and DTCA show moderate correlation, suggesting some coordinated investment s
trategies. While not severe, this relationship should be monitored in regression diagnostic
s.")
```

```
[EDA] Calculating correlation matrix for log-transformed variables...
```

	log_sales	log_detailing	log_dtca
log_sales	1.0000	0.8027	0.6482
log_detailing	0.8027	1.0000	0.5953
log_dtca	0.6482	0.5953	1.0000



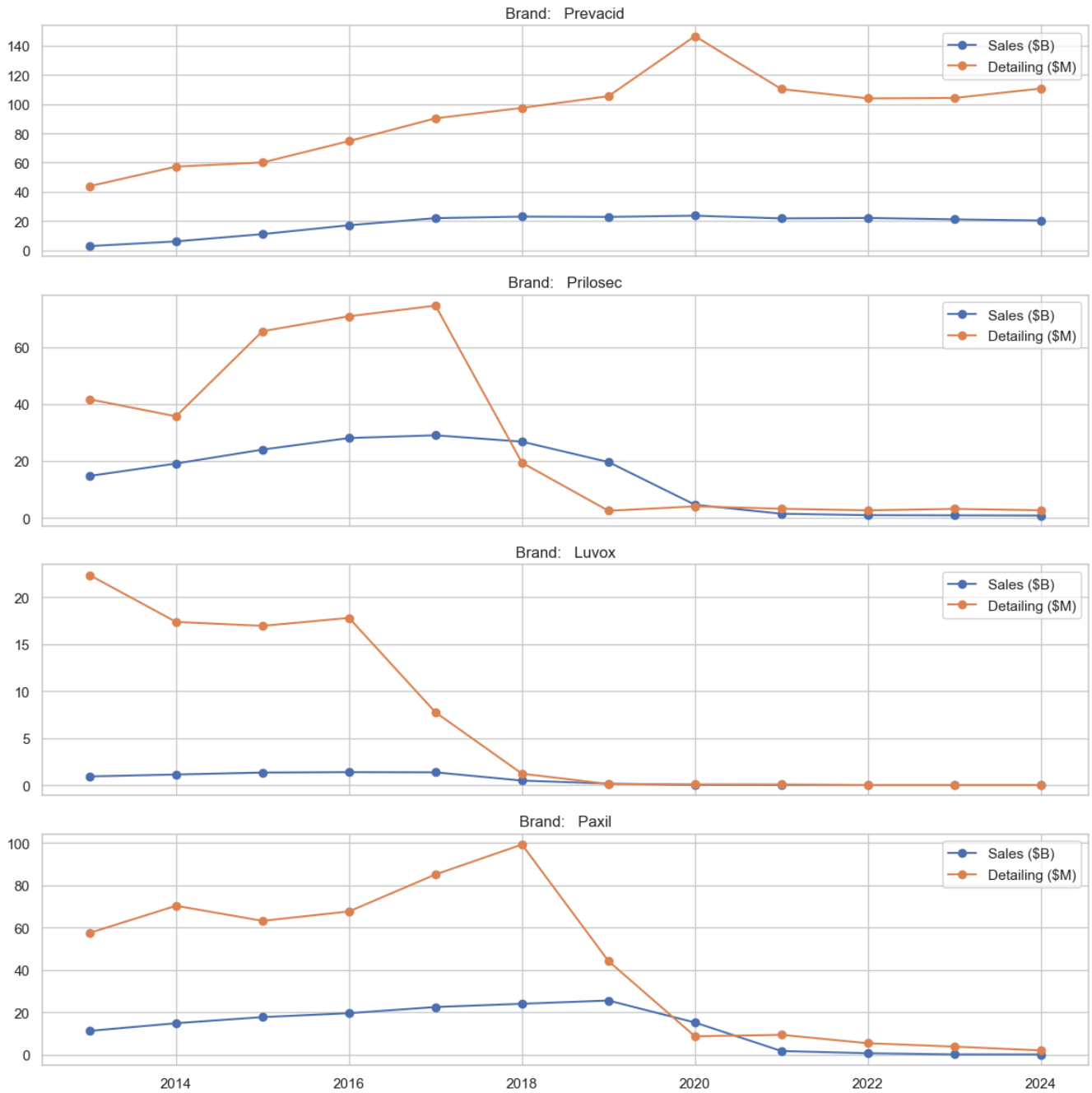

```
[Business Insight]
Detailing and DTCA show moderate correlation, suggesting some coordinated investment strategi
es. While not severe, this relationship should be monitored in regression diagnostics.
```

11. Panel Structure and Within-Brand Dynamics

To isolate marketing response credibly, we need to look beyond cross-sectional size and examine within-brand changes over time.

In [70]:
```
print("\n[EDA] Exploring within-brand trajectories for selected brands...")

sample_agents = df["agent"].drop_duplicates().head(4).tolist()

fig, axes = plt.subplots(len(sample_agents), 1, figsize=(12, 12), sharex=True)

for i, a in enumerate(sample_agents):
    temp = df[df["agent"] == a].sort_values("year")
    axes[i].plot(temp["year"], temp["actual_sales"], marker="o", label="Sales ($B)")
    axes[i].plot(temp["year"], temp["detailing"], marker="o", label="Detailing ($M)")
    axes[i].set_title(f"Brand: {a}")
    axes[i].legend()

plt.tight_layout()
plt.show()

print("\n[Business Insight]")
print("Within-brand variation confirms that changes in marketing over time are associated wit
h changes in sales, which forms the basis for identifying marketing effects in panel model
s.")
print("That is the variation panel-style parametric models exploit for identification.")
```

```
[EDA] Exploring within-brand trajectories for selected brands...
```

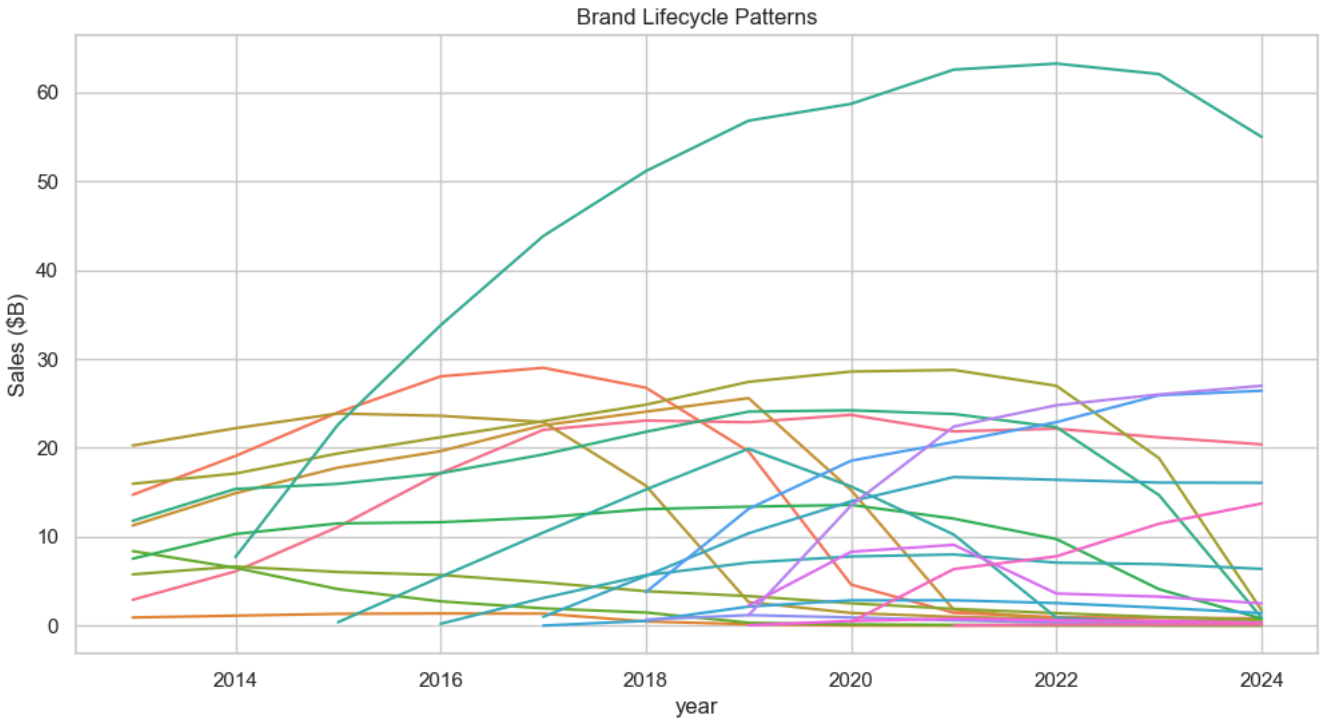



```
[Business Insight]
Within-brand variation confirms that changes in marketing over time are associated with chang
es in sales, which forms the basis for identifying marketing effects in panel models.
That is the variation panel-style parametric models exploit for identification.
```

In [71]:
```
print("\n[EDA] Inferring brand lifecycle patterns...")

agent_lifecycle = (
    df.groupby("agent")
    .agg(
        start_year=("year", "min"),
        end_year=("year", "max"),
        avg_sales=("actual_sales", "mean")
    )
    .sort_values("start_year")
)

display(agent_lifecycle.head(15))

plt.figure(figsize=(12, 6))
sns.lineplot(data=df, x="year", y="actual_sales", hue="agent", legend=False)
plt.title("Brand Lifecycle Patterns")
plt.ylabel("Sales ($B)")
plt.show()

print("\n[Business Insight]")
print("Brands exhibit distinct lifecycle patterns, including entry, growth, and decline, whic
h influence both sales levels and observed marketing effectiveness.")
print("Lifecycle effects may also shape the apparent return to marketing spend.")
```

```
[EDA] Inferring brand lifecycle patterns...
```

agent	start_year	end_year	avg_sales
Zoloft	2013	2024	21.1441
Prozac	2013	2024	11.2791
Prilosec	2013	2024	14.1557
Lescol	2013	2024	3.6356
Prevacid	2013	2024	17.8747
Pravachol	2013	2024	9.9861
Luvox	2013	2024	0.5595
Mevacor	2013	2024	2.1499
Zocor	2013	2024	17.6050
Paxil	2013	2024	12.8184
Lipitor	2014	2024	47.0275
Celexa	2015	2024	7.8978
Aciphex	2016	2024	5.8032
Lescol XL	2017	2024	1.7925
Protonix	2017	2024	12.0249




```
[Business Insight]
Brands exhibit distinct lifecycle patterns, including entry, growth, and decline, which influ
ence both sales levels and observed marketing effectiveness.
Lifecycle effects may also shape the apparent return to marketing spend.
```

12. Outliers and High-Leverage Observations

Before modeling, we identify whether a few brands or campaigns are likely to dominate results.

In [72]:
```
print("\n[EDA] Reviewing highest-leverage observations...")

print("Top 5 by sales:")
display(df.nlargest(5, "actual_sales")[["class", "agent", "year", "actual_sales", "detailin
g", "direct_to_consumer_ad"]])

print("Top 5 by detailing:")
display(df.nlargest(5, "detailing")[["class", "agent", "year", "actual_sales", "detailing",
"direct_to_consumer_ad"]])

print("Top 5 by DTCA:")
display(df.nlargest(5, "direct_to_consumer_ad")[["class", "agent", "year", "actual_sales", "d
etailing", "direct_to_consumer_ad"]])

print("\n[Business Insight]")
print("These observations are not errors; they represent structurally important brands or cam
paigns.")
print("However, they may exert high leverage on pooled model estimates, so robustness checks
will be prudent later.")
```

```
[EDA] Reviewing highest-leverage observations...
Top 5 by sales:
```

	class	agent	year	actual_sales	detailing	direct_to_consumer_ad
159	Statin	Lipitor	2022	63.2186	146.4874	93.4800
137	Statin	Lipitor	2021	62.5406	152.5107	110.7082
181	Statin	Lipitor	2023	62.0389	197.2503	142.6181
115	Statin	Lipitor	2020	58.6918	139.6403	108.3209
94	Statin	Lipitor	2019	56.7952	99.7099	79.1343

```
Top 5 by detailing:
```

	class	agent	year	actual_sales	detailing	direct_to_consumer_ad
191	SSRI	Lexapro	2024	26.9766	243.3303	0.5963
125	SSRI	Lexapro	2021	22.3924	223.0231	0.2595
169	SSRI	Lexapro	2023	26.0019	215.6069	2.4667
203	Statin	Lipitor	2024	54.9727	203.9765	183.0804
104	SSRI	Lexapro	2020	13.5176	202.9746	0.0000

```
Top 5 by DTCA:
```

	class	agent	year	actual_sales	detailing	direct_to_consumer_ad
120	PPI	Nexium	2021	20.6499	168.0683	240.3550
142	PPI	Nexium	2022	22.8833	172.8186	225.8439
99	PPI	Nexium	2020	18.5523	160.8279	215.2616
134	Statin	Crestor	2021	6.3594	200.9031	213.3158
79	PPI	Nexium	2019	13.1240	145.5416	192.2922

```
[Business Insight]
These observations are not errors; they represent structurally important brands or campaigns.
However, they may exert high leverage on pooled model estimates, so robustness checks will be
prudent later.
```

13. EDA Takeaways

In [ ]:
```
print("\n[EDA] Summarizing final business takeaways...")

print("""
1. The portfolio is highly concentrated, with a small number of blockbuster brands driving ov
er half of total revenue.
2. Sales exhibit a clear lifecycle pattern, rising through mid-period before declining, indic
ating that time dynamics must be explicitly modeled.
3. Marketing investments are not uniformly allocated — both channel mix and spend levels vary
significantly across classes and brands.
4. Simple efficiency ratios suggest differences across channels, but are unstable due to stru
ctural zeros and should not be interpreted causally.
5. There is consistent evidence of diminishing returns, particularly in detailing, supporting
nonlinear or log-transformed specifications.
6. Both class and brand identity materially influence baseline sales and likely response to m
arketing, justifying their inclusion as structural controls.
7. Nearly half of observations have zero DTCA, indicating a distinct physician-driven marketi
ng regime that must be modeled separately.
8. The dataset is an unbalanced panel, requiring careful time-based validation to avoid bias
from brand entry and exit.
9. Log transformations substantially improve distributional properties and linearity, enablin
g elasticity-based interpretation of model coefficients.
10. Overall, the evidence supports a log-log parametric model with class, brand, and time con
trols, and a potential regime distinction for DTCA activity.
""")
```

Next Steps for Parametric Modeling

This EDA has already prepared the key features needed for the next stage:

- `log_sales`
- `log_detailing`
- `log_dtca`
- `dtca_zero`

Recommended first parametric baseline

[\log(1 + Sales) \sim \log(1 + Detailing) + \log(1 + DTCA) + Class + Agent + Year]

Extensions suggested by the EDA

- class × channel interaction terms
- `dtca_zero` as a binary regime indicator
- nonlinear response checks if residual diagnostics suggest misfit
- robustness checks for high-leverage brands

In [ ]:
```
print("\n[EDA] Saving processed dataset for the next modeling stage...")

df.to_csv("data/processed_data.csv", index=False)

print("[EDA] Saved processed file: data/processed_data.csv")
```

In [ ]: